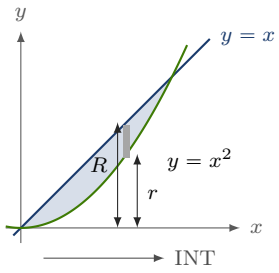
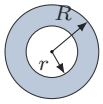
L4.2b Washers

Washers and rotating about other lines — what you need to know

WASHERS





$$\boxed{A = \pi R^2 - \pi r^2}$$

R = outer edge, r = inner edge

$V = \int_a^b A(x)\,dx$ or $V = \int_c^d A(y)\,dy$

(dy when the slices are perpendicular to the y-axis)

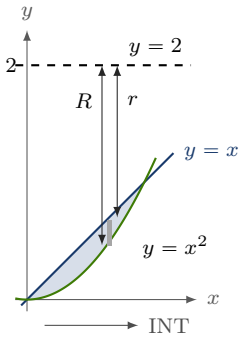
A region that touches the axis gives a disk, the washer with $r = 0$: $A = \pi(\text{radius})^2$.

"the disk out to the outer edge, minus the disk out to the inner edge"

Ex. The region enclosed by $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume.
The curves meet at $x = 0$ and $x = 1$. At $x = 0.5$, $x = 0.5 > 0.25 = x^2$, so $R(x) = x$ and $r(x) = x^2$.

$$A(x) = \pi x^2 - \pi x^4 \qquad V = \pi \int_0^1 x^2\,dx - \pi \int_0^1 x^4\,dx = \frac{\pi}{3} - \frac{\pi}{5} = \boxed{\frac{2\pi}{15}}$$

RADIUS FROM A LINE



about $y = k$: radius $= |k - y|$
about $x = k$: radius $= |x - k|$

① Draw the axis of rotation and a strip perpendicular to it.
② R = distance from the axis to the far curve, r = distance to the near curve.
③ $A = \pi R^2 - \pi r^2$, and integrate along the axis.

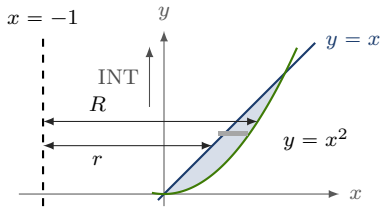
"each radius is still a distance from the axis — measure it from the line, not from 0"

Ex. The same region, rotated about the line $y = 2$. The axis is above the region, so the far curve is $y = x^2$:

$$R(x) = 2 - x^2 \qquad r(x) = 2 - x \qquad A(x) = \pi\left[2 - x^2\right]^2 - \pi(2 - x)^2$$

$$V = \pi \int_0^1 4 - 4x^2 + x^4\,dx - \pi \int_0^1 (2 - x)^2\,dx = \frac{43\pi}{15} - \frac{35\pi}{15} = \boxed{\frac{8\pi}{15}}$$

WASHERS ALONG y



$$\boxed{V = \int_c^d \pi R(y)^2 - \pi r(y)^2 \, dy}$$

About a vertical line, slice perpendicular to it: solve each curve for x, and the limits c and d are **y-coordinates**.

"a vertical axis means horizontal strips, so everything is in terms of y"

Ex. The same region, rotated about the line $x = -1$. Solve: $y = x$ gives $x = y$, and $y = x^2$ gives $x = \sqrt{y}$, for $0 \leqslant y \leqslant 1$. The axis is to the left, so each radius is the curve's x plus 1:

$$R(y) = 1 + \sqrt{y} \qquad r(y) = 1 + y$$

$$V = \pi \int_0^1 (1+\sqrt{y})^2 - (1+y)^2 \, dy = \pi \int_0^1 2\sqrt{y} - y - y^2 \, dy = \pi \left(\frac{4}{3} - \frac{1}{2} - \frac{1}{3} \right) = \boxed{\frac{\pi}{2}}$$

WATCH OUT

✗ ~~$A = \pi(R - r)^2$~~ → $A = \pi R^2 - \pi r^2$ — **square each radius, then subtract**

✗ ~~about $y = 2$, the radius out to $y = x^2$ is x^2~~ → **measure from the axis:** $2 - x^2$

✗ ~~a negative volume~~ → **R and r are swapped — R is the outer edge**

✗ ~~x-limits on a dy integral~~ → **the limits are the y-coordinates of the crossings**